A2 · Unit 5 · WCH15 Transition Metals and Organic Nitrogen Chemistry

Arenes

Benzene has six carbons and three apparent double bonds, yet it refuses to react like an alkene. This topic explains why: its electrons are spread around the whole ring, which makes it unusually stable and steers it towards substitution rather than addition. It also covers the reactions that put new groups onto the ring, and why phenol reacts so much more readily.

The structure of benzene

Organic compounds are either **aliphatic** (straight or branched chains, and rings that are not aromatic) or **aromatic**, containing one or more benzene rings. Compounds containing a benzene ring are called **arenes**.

Benzene, C_6H_6, is a flat hexagonal ring. Each carbon forms σ bonds to two other carbons and one hydrogen, with bond angles of 120°. That leaves each carbon with one electron in a p orbital at right angles to the ring. The six p orbitals overlap sideways all the way round, forming rings of π electron density above and below the plane: the six π electrons are **delocalised** over all six carbons rather than held in three separate double bonds. The ring is drawn as a hexagon with a circle inside.

Evidence against Kekulé's structure

In 1865 Kekulé proposed a ring of alternating single and double bonds, cyclohexa-1,3,5-triene. Three pieces of evidence show that this is wrong:

- **Bond lengths**: X-ray diffraction shows all six C–C bonds are the same length, 0.139 nm, between a C–C single bond (0.154 nm) and a C=C double bond (0.134 nm). Kekulé's structure would have three of each.
- **Enthalpy of hydrogenation**: hydrogenating cyclohexene, with one C=C, releases 120 kJ mol^{-1}, so three C=C bonds should release 3 × 120 = 360 kJ mol^{-1}. Benzene actually releases only 208 kJ mol^{-1}. Benzene is **152 kJ mol^{-1} more stable** than Kekulé's structure: this is the **delocalisation energy**.
- **Reactivity**: alkenes decolourise bromine water quickly by addition, but benzene does not react with bromine water. Addition would break up the stable delocalised system.

Naming aromatic compounds

- Simple derivatives are named with benzene as the parent: methylbenzene, chlorobenzene, nitrobenzene, benzenecarboxylic acid (C_6H_5COOH), benzaldehyde (C_6H_5CHO).
- With two or more substituents, number the ring to give the lowest numbers, and list different groups alphabetically: 1,3-dimethylbenzene, 1-chloro-4-methylbenzene, 2-methyl-1,3,5-trinitrobenzene (TNT).
- When the ring is attached to a more important group, it is called the **phenyl** group, C_6H_5–: phenylamine ($C_6H_5NH_2$), phenylethene, phenylethanone ($C_6H_5COCH_3$), phenyl ethanoate.

- **Toxicity**: benzene is a carcinogen, so methylbenzene is used instead in school laboratories.

Combustion and hydrogenation

- **Combustion**: $C_6H_6 + 7½O_2 \rightarrow 6CO_2 + 3H_2O$. Arenes burn with a very **smoky, sooty flame** because their high carbon-to-hydrogen ratio leaves unburnt carbon.
- **Hydrogenation**: with hydrogen over a nickel catalyst at about 200 °C and 30 atm, benzene adds three molecules of H_2 to give cyclohexane. This is one of the few addition reactions of benzene, and it needs far harsher conditions than for an alkene.

Electrophilic substitution

The delocalised π electrons give benzene a region of high electron density that attracts **electrophiles**. But addition would destroy the delocalisation, so instead benzene undergoes **electrophilic substitution**: one hydrogen is replaced and the stable ring is restored. Every mechanism follows the same three stages:

1. **Making the electrophile**: a catalyst turns the reagent into a positive ion, E^+.
2. **Attack**: a pair of π electrons from the ring forms a bond to E^+. This gives a positively charged intermediate in which the delocalisation covers only five carbons (drawn as a horseshoe from carbon 2 to carbon 6), with the carbon that was attacked holding both H and E.
3. **Loss of H^+**: the electrons of the C–H bond return to the ring, restoring the full delocalised system, and H^+ is released, usually regenerating the catalyst.

Reaction	Reagents and conditions	Electrophile and how it forms	Product
bromination (halogenation)	bromine with an iron(III) bromide catalyst (or iron filings, which form $FeBr_3$); room temperature. Chlorine with $AlCl_3$ or $FeCl_3$ works the same way.	Br^+: $Br_2 + FeBr_3 \rightarrow Br^+ + FeBr_4^-$; then $H^+ + FeBr_4^- \rightarrow FeBr_3 + HBr$	bromobenzene + HBr
nitration	concentrated nitric acid and concentrated sulfuric acid, below about 55 °C (higher temperatures add more nitro groups)	NO_2^+, the nitronium ion: $HNO_3 + 2H_2SO_4 \rightarrow NO_2^+ + H_3O^+ + 2HSO_4^-$; then $H^+ + HSO_4^- \rightarrow H_2SO_4$	nitrobenzene
sulfonation	fuming sulfuric acid (SO_3 in concentrated H_2SO_4), heated under reflux	SO_3, whose sulfur carries a large δ+ charge from its three oxygens	benzenesulfonic acid, $C_6H_5SO_3H$ (used for detergents, dyes and drugs)

Friedel–Crafts alkylation	a halogenoalkane with an anhydrous aluminium chloride catalyst, heated under reflux	R^+: $CH_3CH_2Cl + AlCl_3 \rightarrow CH_3CH_2^+ + AlCl_4^-$; then $H^+ + AlCl_4^- \rightarrow AlCl_3 + HCl$	ethylbenzene + HCl
Friedel–Crafts acylation	an acyl chloride with an anhydrous $AlCl_3$ catalyst, heated under reflux at about 50 °C	RCO^+, the acylium ion: $CH_3COCl + AlCl_3 \rightarrow CH_3CO^+ + AlCl_4^-$	phenylethanone, $C_6H_5COCH_3$, + HCl

EXAM TIP

Friedel–Crafts reactions matter in synthesis because they form a new **carbon–carbon bond** to the ring. Nitration is the first step towards phenylamine, used to make dyes, and towards explosives such as TNT.

COMMON MISTAKE

The aluminium chloride must be **anhydrous**: water would react with it and destroy the catalyst. In the mechanism, the curly arrow for the attack starts from the **circle** (the delocalised electrons), not from a single bond.

How side groups change the ring

Electron-releasing groups such as alkyl groups, –OH and $–NH_2$ push electron density into the ring, making it more attractive to electrophiles. Methylbenzene therefore nitrates more readily than benzene, under milder conditions.

When an atom with a lone pair (O, N or Cl) is bonded directly to the ring, its lone pair overlaps with the π system and becomes partly delocalised into it. This changes both the ring and the side group:

- **Chlorobenzene**: the C–Cl bond gains some double-bond character and becomes stronger, so chlorobenzene does not undergo the substitution and elimination reactions of halogenoalkanes; the electron-rich ring also repels nucleophiles.
- **Phenol**: the C–O bond becomes stronger and the O–H bond weaker, so phenol is more acidic than an alcohol and is not oxidised like one.
- **Phenylamine**: the nitrogen lone pair is partly delocalised, so it is less available to accept a proton, and phenylamine is a much weaker base than aliphatic amines (see Topic 19).

Phenol

In **phenol**, C_6H_5OH, the –OH group is bonded directly to the ring. ($C_6H_5CH_2OH$, with a CH_2 in between, is an ordinary alcohol, phenylmethanol.)

Phenol as an acid

The phenoxide ion is stabilised because its negative charge is partly delocalised into the ring, so phenol is a **weak acid**: stronger than alcohols and water, but weaker than carboxylic acids.

Reagent	Ethanol	Phenol	Ethanoic acid
sodium	reacts: H_2	reacts: $C_6H_5OH + Na \rightarrow C_6H_5O^-Na^+ + ½H_2$	reacts: H_2
sodium hydroxide	no reaction	reacts: $C_6H_5OH + NaOH \rightarrow C_6H_5O^-Na^+ + H_2O$ (the solid dissolves, as sodium phenoxide is soluble)	reacts
sodium carbonate	no reaction	**no reaction**: phenol is too weak an acid	fizzes: CO_2

The sodium carbonate test therefore tells a phenol from a carboxylic acid.

Phenol with bromine water

Phenol decolourises bromine water at room temperature, with no catalyst, giving a **white precipitate** of 2,4,6-tribromophenol:

$$C_6H_5OH + 3Br_2 \rightarrow C_6H_2Br_3OH + 3HBr$$

The oxygen lone pair delocalised into the ring raises its electron density so much that it **polarises** bromine molecules directly, without needing a halogen carrier, and three hydrogens are substituted rather than one. Benzene needs $FeBr_3$ and pure bromine, and substitutes only once. Phenols are used to make plastics, resins, antiseptics and disinfectants.

Quick checks

QUICK CHECK

Which observation is the best evidence that benzene's six C–C bonds are identical?

- benzene burns with a smoky flame
- X-ray diffraction shows all six C–C bonds are 0.139 nm long
- benzene is a liquid at room temperature
- benzene reacts with hydrogen over nickel

Kekulé's structure would have three short C=C and three long C–C bonds. One intermediate length shows the π electrons are spread evenly round the ring.

QUICK CHECK

The enthalpy of hydrogenation of cyclohexene is −120 kJ mol^{-1} and of benzene −208 kJ mol^{-1}. What is the delocalisation energy of benzene?

- 88 kJ mol^{-1}
- 208 kJ mol^{-1}
- 152 kJ mol^{-1}
- 328 kJ mol^{-1}

Three C=C bonds would release 3 × 120 = 360 kJ mol^{-1}; benzene releases 208, so it is 360 − 208 = 152 kJ mol^{-1} more stable. 88 is 208 − 120, which compares with only one double bond.

QUICK CHECK

Why does benzene undergo substitution rather than addition with electrophiles?

- benzene has no double bonds at all
- electrophiles cannot attack benzene
- substitution keeps the stable delocalised π system, which addition would destroy
- addition products are too unstable to isolate

Losing H^+ at the end restores the full ring of delocalised electrons and its delocalisation energy.

QUICK CHECK

What is the electrophile in the nitration of benzene, and how is it formed?

- NO_3^-, from nitric acid losing H^+
- NO_2^+, from $HNO_3 + 2H_2SO_4 \rightarrow NO_2^+ + H_3O^+ + 2HSO_4^-$
- H^+, from sulfuric acid
- NO_2, a radical made by heating

Sulfuric acid, the stronger acid, protonates nitric acid, which then loses water to give the nitronium ion. Sulfuric acid is regenerated at the end.

QUICK CHECK

Which reagents would convert benzene into phenylethanone, $C_6H_5COCH_3$?

- ethanol and concentrated sulfuric acid
- chloroethane and $AlCl_3$
- ethanoyl chloride and anhydrous $AlCl_3$
- ethanoic acid and $FeBr_3$

Friedel–Crafts acylation: $AlCl_3$ forms the acylium ion CH_3CO^+ from ethanoyl chloride. Chloroethane would give ethylbenzene by alkylation.

QUICK CHECK

How can you distinguish phenol from ethanoic acid?

- add sodium: only phenol fizzes
- add sodium hydroxide: only phenol reacts
- add sodium carbonate: only ethanoic acid fizzes
- measure the pH: both are neutral

Both react with sodium and with sodium hydroxide, but phenol is too weak an acid to release CO_2 from a carbonate.

QUICK CHECK

Why does phenol react with bromine water without a catalyst, while benzene does not?

- phenol is a stronger acid than benzene
- a lone pair on oxygen is delocalised into the ring, raising its electron density enough to polarise Br_2
- the OH group is an electrophile
- phenol has a double bond outside the ring

The extra electron density makes the ring so attractive to electrophiles that Br_2 is polarised directly, and substitution happens at three positions to give 2,4,6-tribromophenol.

QUICK CHECK

In the mechanism of electrophilic substitution, what does the "horseshoe" in the intermediate show?

- the electrophile is attached to every carbon
- the remaining delocalised electrons and the positive charge are spread over five carbons
- the ring has opened
- the hydrogen has already been lost

Two of the six π electrons have formed the bond to the electrophile, so the other four, with the positive charge, are delocalised over the remaining five carbons. The sixth carbon carries both H and E.